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Editor-in-Chief
IEEE Access
IEEE

Dear Editor-in-Chief,

I am writing to submit our manuscript entitled, “**XAI-SDN: Explainable Artificial Intelligence for Real-Time DDoS Detection in Software-Defined Networks**,” for consideration for publication in *IEEE Access*.

In this paper, we address a critical challenge in modern network security: the lack of transparency in high-performance, ML-driven intrusion detection systems. While Software-Defined Networking (SDN) enables unprecedented network control, its centralized architecture is highly vulnerable to volumetric DDoS attacks. Existing machine learning solutions achieve high detection accuracy but operate as opaque “black boxes,” making it difficult for security operators to trust or verify automated mitigation actions during an ongoing attack.

To bridge this gap, we introduce XAI-SDN, a real-time detection framework that combines an $\mathcal{O}(1)$ rolling entropy engine with a Random Forest classifier and TreeSHAP attribution. Our extensive evaluation on the CIC-DDoS2019 dataset demonstrates that XAI-SDN achieves near-perfect detection accuracy (99.99%) and macro F1 (99.96%) while maintaining a latency footprint of just 0.5 milliseconds per alert. Crucially, our system provides precise, per-decision explanations that empower network operators to understand exactly which traffic features triggered an alert, without sacrificing the throughput required for production environments.

We believe that our work aligns perfectly with the scope of *IEEE Access*, as it presents a practical, rigorously evaluated solution to a pressing problem at the intersection of network security and explainable AI. The operational perspective we provide—specifically regarding the computational cost of real-time explainability and the sensitivity of false positive rates to evaluation methodologies—will be of immediate interest and high value to the IEEE Access readership.

This manuscript is original, has not been published previously, and is not currently under consideration for publication elsewhere. All authors have reviewed and approved the final version of the manuscript.

Thank you for your time and for considering our work. We look forward to the opportunity to

receive feedback from the reviewers.

Sincerely,

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